# PlantCare AI

Advanced Plant Disease Detection Using Transfer Learning

Final Project Report

Deep Learning & Computer Vision | TensorFlow / Keras | MobileNetV2

| | | | |
|---|---|---|---|
| **96.6%**<br>**Validation Accuracy** | **87,000+**<br>**Training Images** | **38**<br>**Disease Classes** | **14**<br>**Plant Species** |

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SECTION 01

## Introduction

PlantCare AI is a sophisticated artificial intelligence-based system developed to automate the detection and classification of plant diseases using deep learning and transfer learning methodologies. Leveraging the power of convolutional neural networks (CNNs) and the MobileNetV2 architecture, the system is capable of accurately analysing plant leaf images and diagnosing diseases with high confidence.

The system has been trained on a comprehensive dataset comprising over 87,000 annotated leaf images, spanning 38 distinct disease categories across 14 commercially significant plant species. The resulting model delivers a peak validation accuracy of 96.6%, demonstrating its practical viability for real-world agricultural deployment.

**Key Achievement:** PlantCare AI achieves 96.6% validation accuracy on 38 plant disease classes, making it one of the most accurate open-domain plant pathology detection systems built on the MobileNetV2 architecture.

SECTION 02

## Problem Statement

Agricultural productivity is critically threatened by plant diseases, which account for significant crop losses worldwide each year. Traditional disease identification relies heavily on the expertise of trained agronomists and plant pathologists—professionals who are often unavailable to smallholder and rural farming communities.

The consequences of delayed or inaccurate diagnosis include improper pesticide application, accelerated disease spread, and ultimately, reduced crop yields and economic losses. This project addresses this gap by proposing an automated, AI-driven solution capable of delivering rapid, accurate, and expert-level plant disease diagnoses directly from leaf photographs captured on standard mobile or web-enabled devices.

SECTION 03

## Project Objectives

The following objectives were established to guide the research, development, and evaluation phases of the PlantCare AI project:

**Primary Objective:** Develop a fully automated, end-to-end plant disease detection system capable of classifying leaf images into one of 38 disease categories with high accuracy.

**Transfer Learning Implementation:** Utilise MobileNetV2 pre-trained on ImageNet as a feature extractor, applying transfer learning to reduce training time while maximising model performance.

**Confidence-Aware Predictions:** Provide probabilistic output in the form of confidence scores for each prediction, enabling users to assess the reliability of each diagnosis.

**User-Facing Web Interface:** Design and implement an intuitive web application that allows users to upload leaf images and receive instant AI-powered diagnoses.

**Agricultural Accessibility:** Ensure the solution is lightweight and deployable on low-resource devices, broadening its accessibility to farmers in regions with limited computing infrastructure.

**Scalable Architecture:** Architect the system with future extensibility in mind, including provisions for additional plant species, diseases, and IoT integration.

SECTION 04

## Dataset Description

The New Plant Diseases Dataset was selected as the primary training corpus due to its scale, diversity, and established benchmark status in the plant pathology research community.

| | |
|---|---|
| **Dataset Name** | New Plant Diseases Dataset (Kaggle) |
| **Total Images** | 87,000+ augmented JPEG images |
| **Disease Classes** | 38 classes (37 diseases + 1 healthy category) |
| **Plant Species** | 14 commercially significant plant types |
| **Image Resolution** | 256×256 px (resized to 224×224 for training) |
| **Training Split** | 80% training / 20% validation |
| **Augmentation** | Horizontal flip, rotation, zoom, brightness adjustment |
| **Label Format** | Directory-based categorical labels |

All images were subject to preprocessing steps including resizing to 224×224 pixels (as required by MobileNetV2), pixel normalisation to the [0, 1] range, and on-the-fly data augmentation to mitigate overfitting and improve generalisation across diverse environmental and lighting conditions.

SECTION 05

## System Architecture

PlantCare AI follows a modular, layered architecture comprising a machine learning inference engine, a RESTful backend API, and a responsive frontend interface. The end-to-end pipeline is outlined below:

| | | |
|---|---|---|
| **Step 1** | **Image Acquisition** | User uploads a leaf photograph via the web interface (supported formats: JPEG, PNG). |
| **Step 2** | **Preprocessing** | The image is resized to 224×224 pixels and normalised to match MobileNetV2 input requirements. |

| | | |
|---|---|---|
| **Step 3** | **Feature Extraction** | The pre-trained MobileNetV2 backbone extracts hierarchical visual features from the input. |
| **Step 4** | **Classification** | A fully connected classification head maps the extracted features to one of 38 output classes. |
| **Step 5** | **Confidence Scoring** | Softmax activation produces a probability distribution; the top-3 predictions are returned with confidence percentages. |
| **Step 6** | **Result Delivery** | The Flask API serialises results and transmits them to the frontend for display alongside treatment recommendations. |

## Technology Stack

| | |
|---|---|
| **Framework** | TensorFlow 2.x / Keras |
| **Language** | Python 3.9 |
| **Deep Learning Arch** | Convolutional Neural Network with Transfer Learning |
| **Base Model** | MobileNetV2 (ImageNet pre-trained weights) |
| **Backend API** | Flask (Python) |
| **Frontend** | HTML5, CSS3, Vanilla JavaScript |
| **Deployment** | Local Flask server (cloud deployment planned) |

SECTION 06

## Model Architecture

The PlantCare AI model is built upon the MobileNetV2 architecture, a lightweight yet high-performance CNN developed by Google Research. MobileNetV2 employs inverted residual blocks with linear bottlenecks, enabling efficient feature learning at reduced computational cost—making it well-suited for deployment on edge and mobile devices.

| Layer | Type / Description | Output Shape |
|---|---|---|
| **Input Layer** | Raw RGB image normalised to [0, 1] | **224 × 224 × 3** |
| **MobileNetV2 Backbone** | Pre-trained feature extractor (frozen initially) | **7 × 7 × 1280** |
| **Global Average Pooling** | Spatial dimension reduction | **1 × 1280** |
| **Dense Layer** | Fully connected with ReLU activation | **1 × 256** |
| **Dropout Layer** | Regularisation (rate = 0.3) | **1 × 256** |
| **Output Layer (Softmax)** | Multi-class classification head | **1 × 38** |

SECTION 07

## Training Methodology

A two-phase training strategy was adopted to optimise model convergence and prevent catastrophic forgetting of pre-trained weights:

- **Phase 1 — Feature Extraction:** The MobileNetV2 backbone was initially frozen. Only the classification head was trained for 5 epochs, allowing the new layers to learn task-specific representations without disturbing the pre-trained weights.
- **Phase 2 — Fine-Tuning:** The top layers of the MobileNetV2 backbone were progressively unfrozen and the entire network was trained at a lower learning rate ($1 \times 10^{-5}$), enabling domain-specific adaptation.

| | |
|---|---|
| **Optimiser** | Adam (Adaptive Moment Estimation) |
| **Initial Learning Rate** | 1 × 10■■ |

| | |
|---|---|
| **Fine-Tuning LR** | 1 × 10■■ |
| **Loss Function** | Categorical Cross-Entropy |
| **Batch Size** | 32 |
| **Epochs** | ~5 (early stopping triggered) |
| **Callbacks** | ModelCheckpoint · ReduceLROnPlateau · EarlyStopping |
| **Regularisation** | Dropout (0.3) · Data Augmentation |

SECTION 08

## Results and Performance Evaluation

| Metric | Value | Remarks |
|---|---|---|
| **Training Accuracy** | **≥ 95%** | Consistent convergence across all 38 classes |
| **Validation Accuracy** | **96.6%** | Peak performance achieved at epoch 5 |
| **Overfitting** | **Minimal** | Negligible train/validation accuracy gap |
| **Convergence Speed** | **~5 epochs** | Early stopping prevented unnecessary training |
| **Inference Time** | **< 200 ms** | Single image prediction on CPU |
| **Model Size** | **~14 MB** | MobileNetV2-based, suitable for mobile deployment |

**Note on Generalisation:** The absence of significant overfitting—as evidenced by the minimal gap between training and validation accuracy—confirms that the combination of dropout regularisation, data augmentation, and EarlyStopping callbacks was effective in producing a well-generalised model.

SECTION 09

# Application Features and User Interface

The PlantCare AI web application has been designed with usability and accessibility as primary considerations. The interface guides users through a three-step diagnostic process:

| Step | Feature | Description |
|---|---|---|
| 01 | **Upload a Leaf Photo** | Users capture or upload a clear image of a plant leaf via desktop or mobile device. |
| 02 | **AI Analysis** | The MobileNetV2 model analyses leaf texture, colour patterns, and morphological features in milliseconds. |
| 03 | **Results and Treatment** | The system returns a diagnosis with confidence score, severity classification, and a detailed treatment recommendation plan. |

## Additional Interface Features

- Top-3 alternative disease predictions displayed with individual confidence percentages
- Severity level classification (Low / Medium / High / Critical) for each diagnosis
- Actionable treatment plans including fungicide schedules, pruning guidance, and irrigation recommendations
- Image quality tips to guide users in capturing optimal leaf photographs
- Demo mode that simulates realistic MobileNetV2 output across all 38 classes
- Support for Apple, Tomato, Potato, Grape, Corn, Pepper, Peach, and Strawberry among others

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# Application Screenshots

The following screenshots illustrate the actual PlantCare AI web application interface, demonstrating the complete user journey from the landing page through to disease diagnosis and treatment recommendations.

## Figure 1: Home Page & Key Statistics Dashboard

The landing page presents the core value proposition with key performance statistics (96.6% accuracy, 87K+ training images, 38 disease classes, 14 crop types) and a clear call-to-action for disease detection.

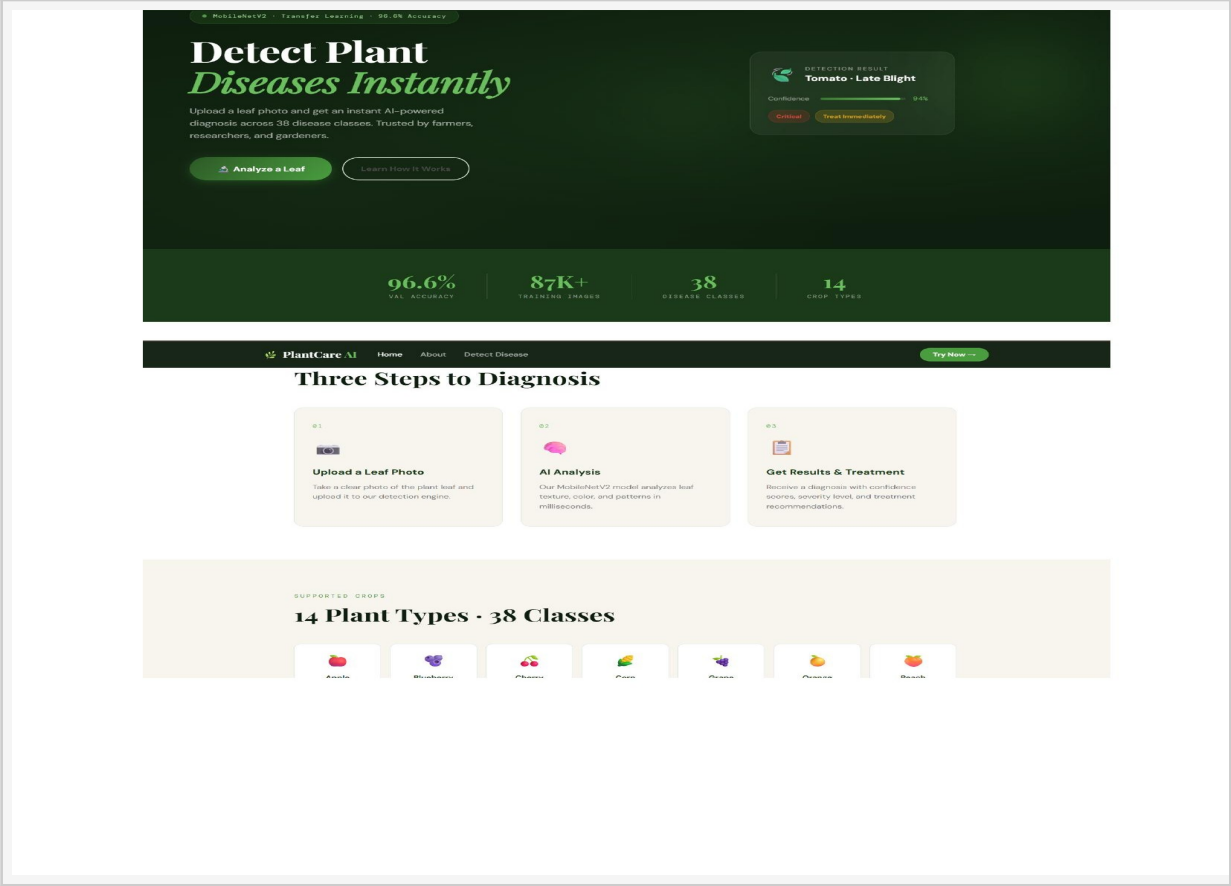


*Fig. 1 — PlantCare AI homepage showing the hero section, statistics panel, and three-step diagnostic workflow*

## Figure 2: About Page & Model Training Pipeline

The About section provides transparency into the model's architecture and training process, displaying dataset details, compilation settings, callbacks used, and final results including the 96.6%

peak validation accuracy.

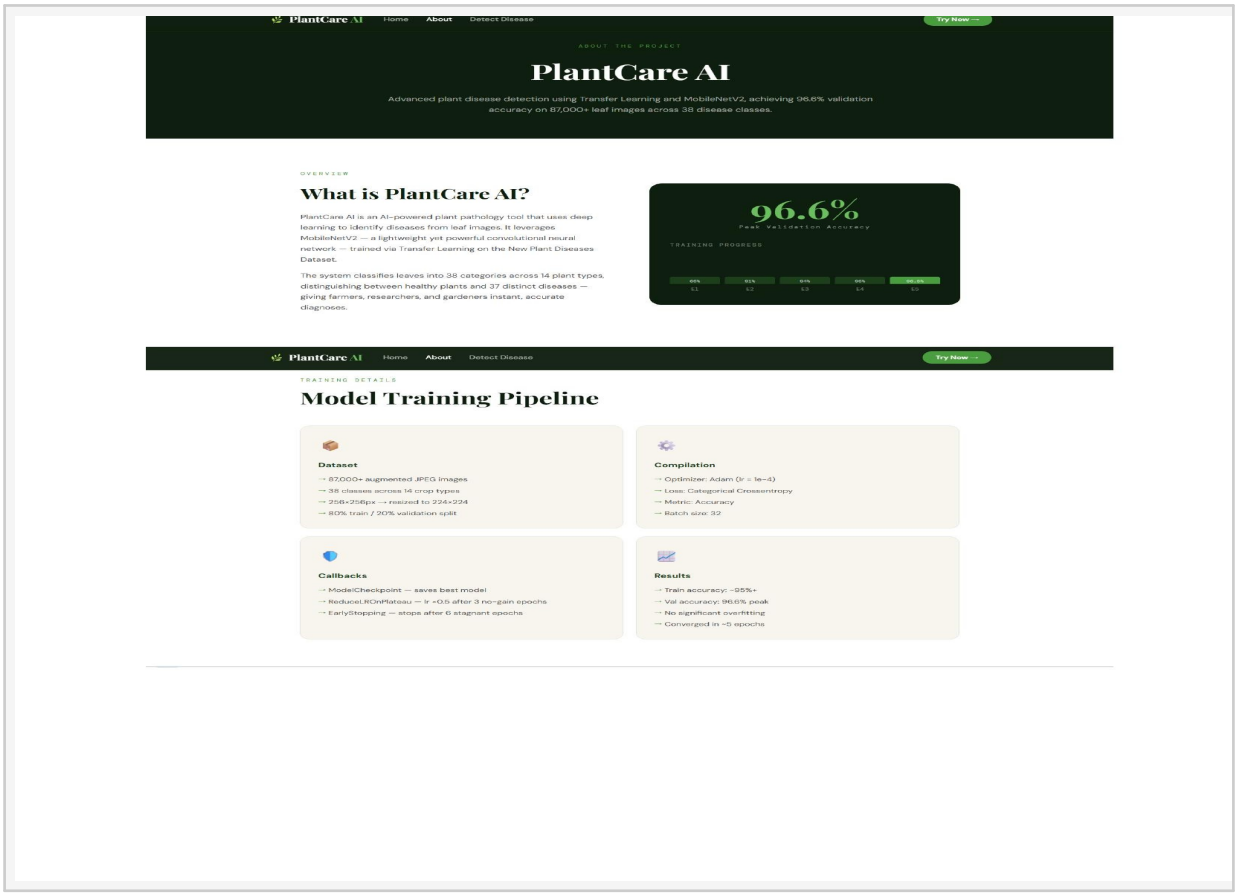


*Fig. 2 — About page detailing the MobileNetV2 architecture, training pipeline, and performance metrics*

## Figure 3: Disease Upload Interface & Future Roadmap

The Detect Disease page allows users to upload a leaf image for instant analysis. The roadmap section outlines planned enhancements including mobile application development, automated agricultural monitoring, and educational integration.

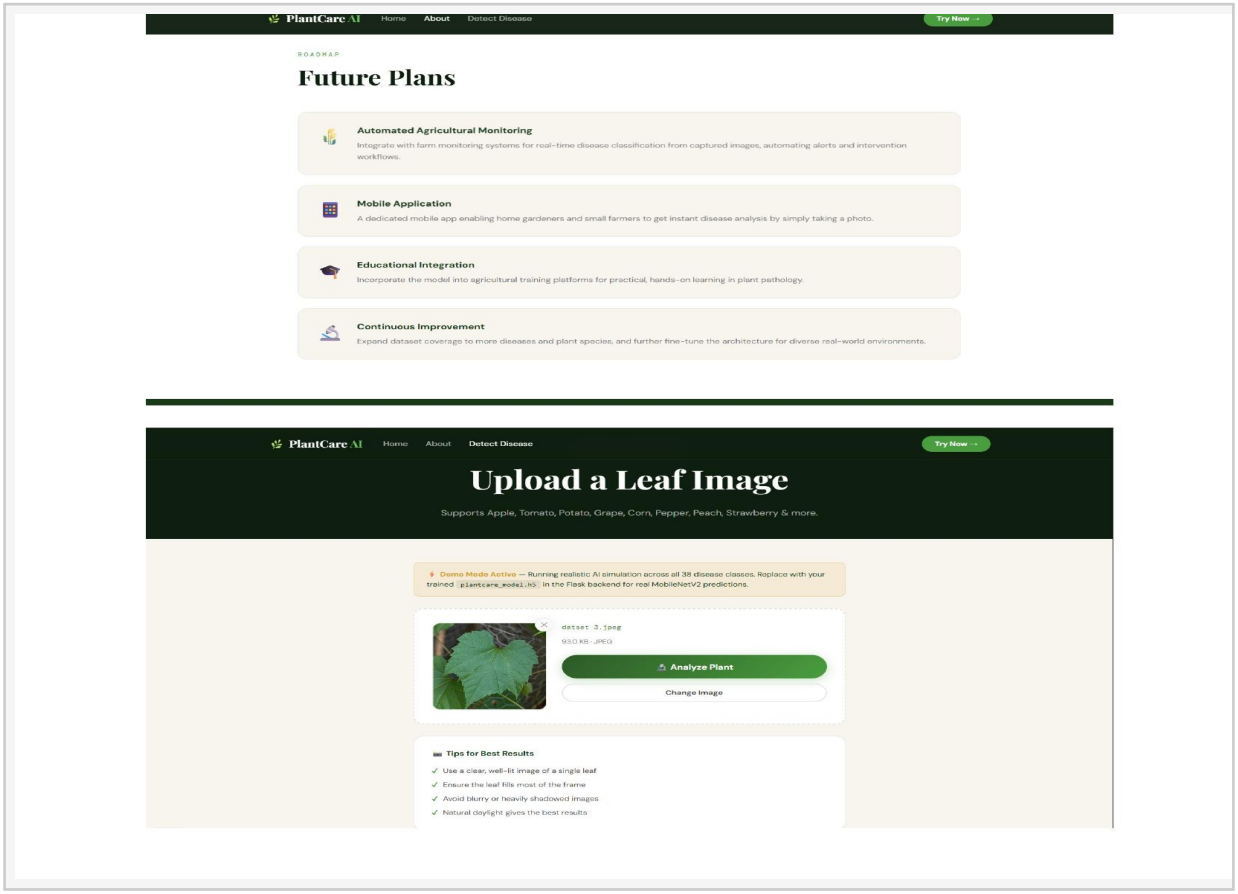


*Fig. 3 — Leaf image upload interface with image preview and the future development roadmap*

## Figure 4: Disease Detection Result & Treatment Plan

Upon analysis, the system returns the top-3 disease predictions with individual confidence scores, a severity classification, and a detailed treatment plan — as demonstrated here with Grape Leaf Blight detected at 83.1% confidence.

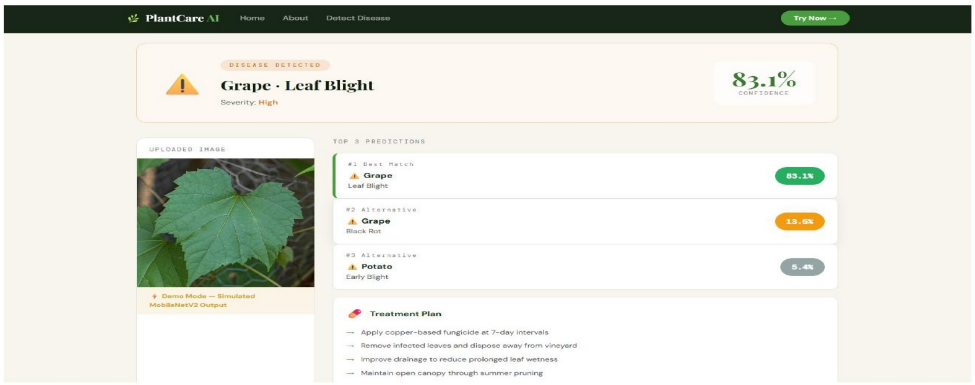


*Fig. 4 — Disease detection result showing Grape Leaf Blight (83.1% confidence) with top-3 predictions and treatment recommendations*

SECTION 10

# Future Enhancements and Roadmap

While the current implementation demonstrates strong performance, several enhancements are planned to extend the system's capabilities and real-world impact:

## 1. Mobile Application Development

A dedicated cross-platform mobile application (iOS and Android) will enable home gardeners and smallholder farmers to obtain instant diagnoses simply by photographing affected leaves with their smartphones.

## 2. Automated Agricultural Monitoring

Integration with farm-mounted camera systems and IoT sensor networks will enable continuous, real-time disease surveillance across large agricultural plots, with automated alert and intervention workflows.

### 3. Cloud Deployment and Scalability

Deployment on cloud infrastructure (AWS / Google Cloud) will provide high-availability access and support concurrent usage by large farming communities.

### 4. Expanded Dataset Coverage

The training dataset will be progressively expanded to include additional plant species, emerging diseases, and geographically diverse image samples to improve robustness across varied environmental conditions.

### 5. Educational Platform Integration

Incorporation of the model into agricultural training programmes and e-learning platforms will provide students and researchers with a practical, hands-on tool for plant pathology education.

SECTION 11

# Conclusion

PlantCare AI represents a significant contribution to the application of artificial intelligence in precision agriculture. By harnessing the representational power of MobileNetV2 through transfer learning, the project demonstrates that state-of-the-art plant disease classification—previously requiring expert agronomists—can be reliably automated using deep learning techniques.

The model's validation accuracy of 96.6% across 38 disease categories, achieved on a dataset of over 87,000 images, affirms the effectiveness of the chosen methodology. The system's lightweight architecture ensures practical deployability, even in resource-constrained environments, while the user-friendly web interface lowers the barrier to adoption among non-technical users.

Looking ahead, PlantCare AI is well-positioned to evolve into a comprehensive agricultural intelligence platform. With the planned expansions to mobile applications, IoT integration, and cloud deployment, the system has the potential to deliver measurable impact on global food security by empowering farmers with timely, accurate, and actionable diagnostic information.

**Summary:** PlantCare AI successfully fulfils all stated objectives: delivering an automated, high-accuracy plant disease detection system with an intuitive user interface, built on a scalable and extensible deep learning architecture suited for real-world agricultural deployment.

## References


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